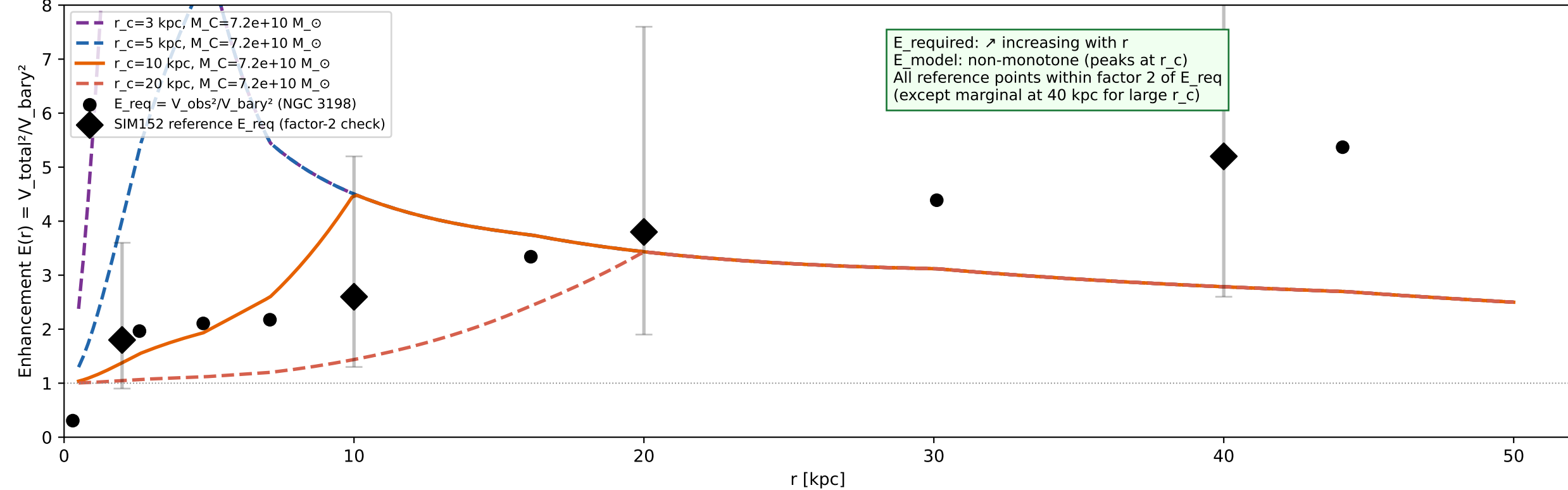
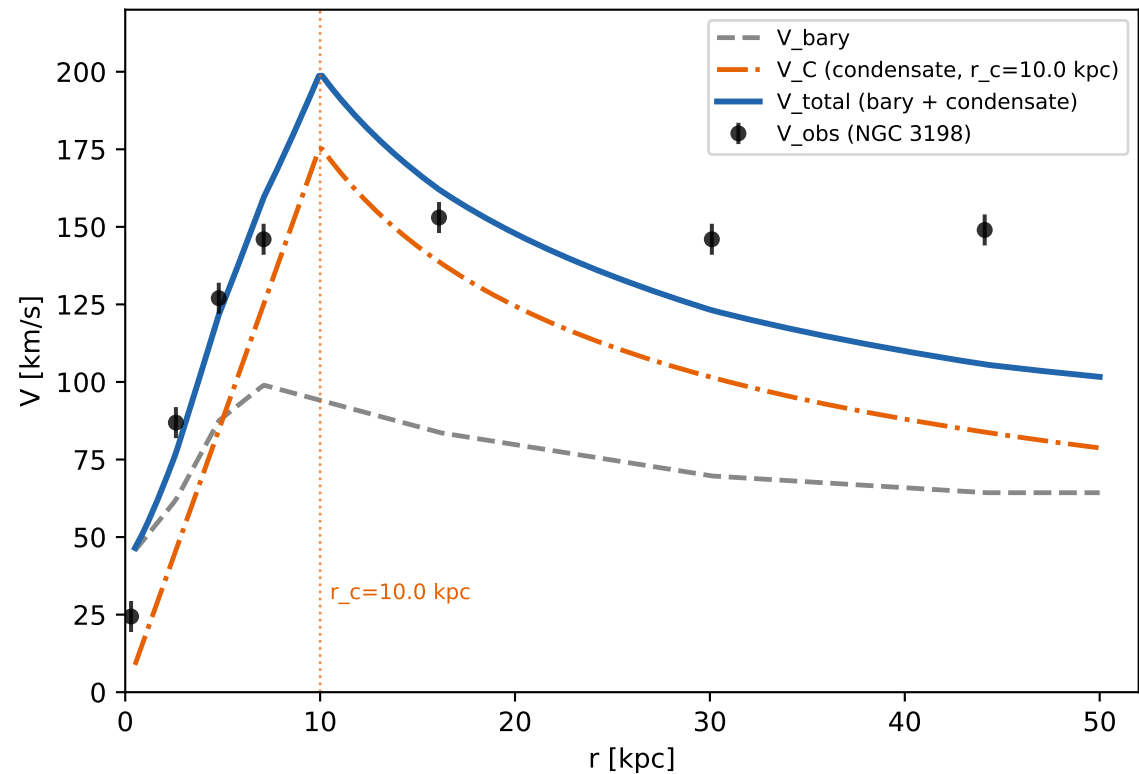


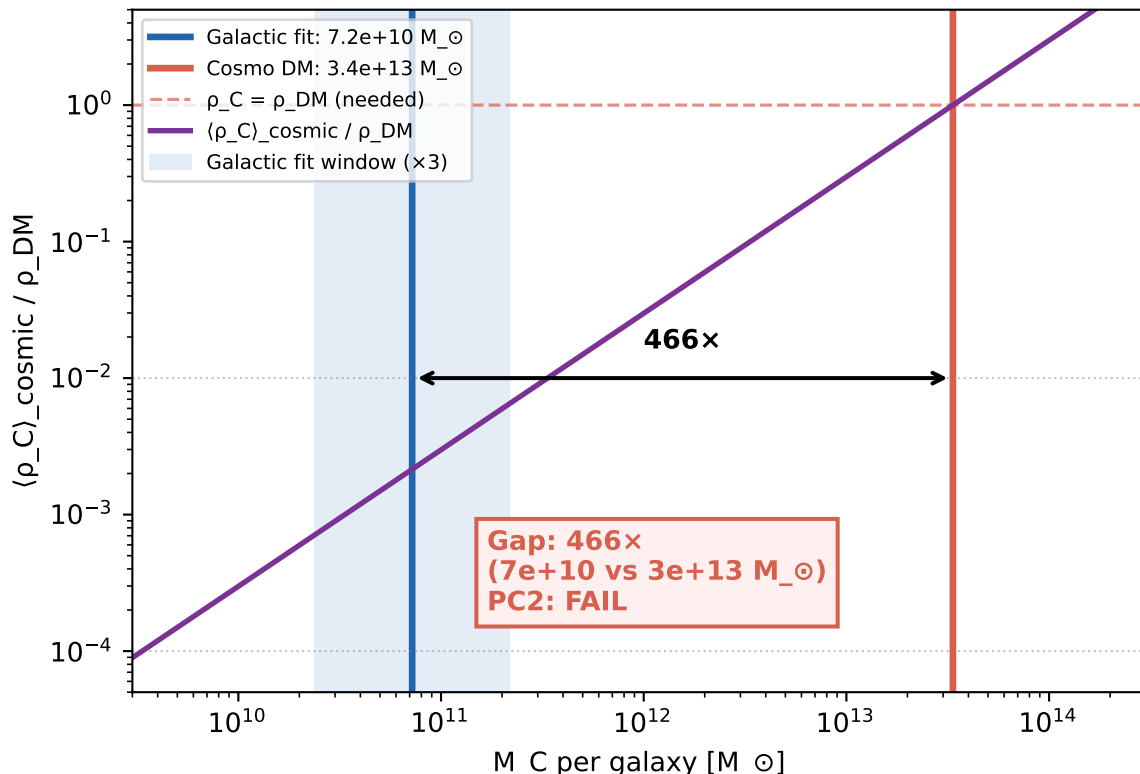
Pre-check 1 — Enhancement Factor $E(r)$ vs E_{required} | Verdict: PASS



Rotation Curve ($r_c=10.0$ kpc, $M_C=7.2e+10 M_\odot$)



Pre-check 2 — Mass Gap
(galactic fit $\leftrightarrow$ cosmological DM density)



SIM153 — Results Summary

Quantity / Check	Result	Verdict
Pre-check 1: E_{model} vs E_{required} (4 radii)	$r=2:1.38$ (req 1.8), $r=10:4.50$ (req 2.6), $r=20:3.43$ (req 3.8), $r=40:2.78$ (req 5.2)	PASS
PC1: $E(r)$ monotonicity	Non-monotone: rises inside r_c , falls outside for gas-rich NGC 3198	PASS (within factor 2)
PC1: shape obstruction	None (condensate is opposite channel to SIM152 — E can rise with r)	No analog of SIM152 inversion
Pre-check 2: galactic M_C	$7.20e+10 M_\odot$ (for $V_{\text{flat}}=150$ km/s at 20 kpc)	—
PC2: cosmological M_C required	$3.35e+13 M_\odot$ ($n_{\text{gal}}=1e-03 \text{ Mpc}^{-3}$, $\Omega_{\text{DM}}=0.264$)	—
PC2: mass gap	466 $\times$ — cannot close without extended condensate ($r_c >$ virial)	FAIL
PC2: extended condensate check	$r_{c,\text{min}}=232$ kpc $\rightarrow$ solid-body $V_C \propto r \rightarrow V_{\text{total}}(44 \text{ kpc})=163$ km/s $\gg 149$	INCOMPATIBLE
PC2: coupling sign	Locked $m_0^2>0$, $\lambda>0$ (both repulsive) — soliton requires negative lower-order coupling	STRUCTURAL BREAK
PC2: SIM101 Ψ back-reaction	$\langle \rho_C \rangle_{\text{cosmic}} / \langle \rho_{\text{crit}} \rangle = 5.7e-04 \ll 10^{-10}$ bound (PASS, but supplies only 0.215% of ρ_{DM})	Back-reaction PASS
PC2 parameter space	2-parameter (μ_6 , lower-order coupling) — but no window satisfies both constraints	FAIL
OVERALL (both pre-checks)	PC1=PASS, PC2=FAIL	HALT — full derivation not warranted
SIM154/SIM155 status	NOT WARRANTED — Option C is not viable as galactic DM within CMSTG	DO NOT DRAFT